

LC 1306 — Jump Game III

Treat each index as a graph node.

Each index has up to 2 edges: $i - a[i]$, $i + a[i]$.

Use DFS + visited to avoid cycles.

$a[i] == 0 \rightarrow$ destination reached.

Check both directions independently.

TC: $O(n)$ | SC: $O(n)$

IMPLEMENTATION (C++)

```
class Solution {
public:
    int n;

    bool dfs(vector<int>& arr, int start, vector<int>& seen) {

        if (seen[start])
            return false;

        seen[start] = 1;
        if (arr[start] == 0)
            return true;

        if (start - arr[start] >= 0) {
            if (dfs(arr, start - arr[start], seen)) {
                return true;
            }
        }

        if (start + arr[start] < n) {
            if (dfs(arr, start + arr[start], seen)) {
                return true;
            }
        }
    }
};
```

```
        }  
    }  
  
    return false;  
}  
bool canReach(vector<int>& arr, int start) {  
  
    n = arr.size();  
    vector<int> seen(n, 0);  
    return dfs(arr, start, seen);  
}  
};
```